Service or Priority-Triggered Inter-Frequency Handover

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1 Service or Priority-Triggered Inter-Frequency Handover Overview

> The feature enables voice service (VoLTE or PTT) on an LTE network with frequency bands where the voice service is prohibited by the frequency assignment authority.

Access Type:	LTE
Feature Identity:	FAJ 121 3087
Value Package Name:	Service-Based Mobility
Value Package Identity:	FAJ 801 0433
Node Type:	Baseband Radio Node
Licensing:	Licensed feature. One license for each node.

Summary

The Service or Priority-Triggered Inter-Frequency Handover feature provides the following benefits:

- It identifies the voice service by its QCI, or by its service type, and moves the UE to a frequency where the service is allowed.
- It improves the possibilities to fulfill the legal requirements on positioning precision for emergency calls. The feature optimizes the OTDOA positioning precision by either of the following:
  - Moving emergency calls to the lowest possible frequency
  - Moving emergency calls to the frequency with the tightest site grid

Additional Information

More information about this feature and related topics can be found in the following documentation:

- Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC); Protocol specification, *3GPP TS 36.331*
- Evolved Universal Terrestrial Radio Access Network (E-UTRAN); S1 Application Protocol (S1AP), *3GPP TS 36.413*

2 Dependencies of Service or Priority-Triggered Inter-Frequency Handover

The feature has conflicts with other features which configure measurements in the UE and trigger handovers.

Table 1 Feature Dependencies

Feature	Relationship	Relationship Scope	Description
Coverage-Triggered Inter-Frequency Handover (FAJ 121 0877)	Prerequisite		This feature initiates a handover to a cell on another frequency.
Multiple Radio Bearers per User (FAJ 121 0549)	Prerequisite		This feature enables more than one bearer for each UE.
Coverage-Triggered Mobility at Setup (FAJ 121 5144)	Related		If the Service or Priority-Triggered Inter-Frequency Handover feature is activated together with the Coverage-Triggered Mobility at Setup feature, the following conditions apply: – If the value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to `QCI`, the Coverage-Triggered Mobility at Setup feature is not triggered. – If the value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to ARP, ARP-related emergency calls do not trigger the Coverage-Triggered Mobility at Setup feature. – Non-emergency calls trigger CS fallback to GERAN or UTRAN.
Device-Based Frequency Filtering (FAJ 121 5154)	Related		The Device-Based Frequency Filtering feature is used to configure a list of unsupported frequencies for specific UEs. These frequencies are excluded from UE configuration messages sent by the eNodeB. Unsupported frequencies are configured using the `UnsupportedFreqProfile.unsupportedFreqList` attribute.
Downlink-Triggered VoLTE Mobility (FAJ 121 5686)	Related		The Downlink-Triggered VoLTE Mobility handovers are disabled if all of the following conditions are fulfilled: – The Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE. – The value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to `QCI`. – The value of the `EUtranCellFDD.servOrPrioIFHoSetupBearer` or the `EUtranCellTDD.servOrPrioIFHoSetupBearer` attribute is set to `false`.
Inter-Frequency OTDOA (FAJ 121 3090)	Related		Used when determining if OTDOA can be used by the UE in the serving cell.
Intra-LTE Inter-Mode Handover (FAJ 121 3042)	Related		Relevant when determining if a cell is usable as a target for handover.
LTE Inter-Frequency Handover to Higher Priority (FAJ 121 5682)	Related		The LTE Inter-Frequency Handover to Higher Priority feature is disabled if the following conditions are fulfilled: – The Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE. – The value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to `QCI`. – The value of the `EUtranCellFDD.servOrPrioIFHoSetupBearer`

Feature	Relationship	Relationship Scope	Description
			or the EUtranCellTDD.servOrPrioIFHoSetupBearer attribute is set to FALSE.
Mission-Critical Push-to-Talk (FAJ 121 4929)	Related		When the Mission-Critical Push-to-Talk feature is activated, all processes and criteria in this feature operation regarding QCI 1 also apply to mission-critical bearers (QCIs 65, 66, and 69).
Multiple Frequency Band Indicators (FAJ 121 3054)	Related		Affects which frequencies can be used for handover.
Multi-Target RRC Connection Re-establishment (FAJ 121 3023)	Related		Support for re-establishment to a different cell than the serving cell.
OTDOA User Plane Location Support (FAJ 121 0862)	Related		Used when determining if OTDOA can be used by the UE in the serving cell.
Service Triggered Mobility (FAJ 121 1747)	Related		If the EUtranCellFDD.spifhoDedicatedReportConfig or the EUtranCellTDD.spifhoDedicatedReportConfig attribute is set to false, the threshold offset and the offset adjustment values that are configured in the Service or Priority-Triggered Inter-Frequency Handover feature affect the event A5 thresholds.
Shared LTE RAN (FAJ 121 0860)	Related		Affects which frequencies can be used for handover.
Subscriber Triggered Mobility (FAJ 121 1788)	Related		Can affect the values for EUtranFreqRelation.voicePrio and EUtranFreqRelation.voicePrio.
Admission-Triggered Offload (FAJ 121 3100)	Related		Event A3 and A5 triggered handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
Coverage-Triggered Inter-Frequency Session Continuity (FAJ 121 0797)	Related		Handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
Inter-Frequency Load Balancing (FAJ 121 3009)	Related		Handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
Inter-Frequency Offload (FAJ 121 3061)	Related		Handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
MBMS Multi-Carrier Support (FAJ 121 3062)	Related		Handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
Mobility Control at Poor Coverage (FAJ 121 3013)	Related		Event A3 and event A5 triggered handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
Uplink-Triggered Inter-Frequency Mobility (FAJ 121 1797)	Related		Event A5 triggered handover disabled to frequency with EUtranFreqRelation.voicePrio set to -1.
Multi-Layer Service-Triggered Mobility (FAJ 121 4124)	Related		If the EUtranCellFDD.spifhoDedicatedReportConfig or the EUtranCellTDD.spifhoDedicatedReportConfig attribute is set to false, the threshold offset and the offset adjustment values that are configured in the Service or Priority-Triggered Inter-Frequency Handover feature affect the event A5 thresholds.
Uplink-Triggered VoLTE Mobility (FAJ 121 5433)	Related		The Uplink-Triggered VoLTE Mobility handovers are disabled if all of the following conditions are fulfilled: – The Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE. – The value of the EUtranCellFDD.servOrPrioTriggeredIFHo or the EUtranCellTDD.servOrPrioTriggeredIFHo attribute is set to QCI. – The value of the EUtranCellFDD.servOrPrioIFHoSetupBearer or the EUtranCellTDD.servOrPrioIFHoSetupBearer attribute is set to FALSE.
Coverage-Triggered CDMA-eHRPD Session Continuity (FAJ 121 0491)	Conflicting		All handovers disabled.
Coverage-Triggered GERAN Session Continuity (FAJ 121 0495)	Conflicting		All handovers disabled.
Coverage-Triggered TD-SCDMA Session Continuity (FAJ 121 3004)	Conflicting		All handovers disabled.

Feature	Relationship	Relationship Scope	Description
Coverage-Triggered WCDMA Session Continuity (FAJ 121 0493)	Conflicting		All handovers disabled.
Intra-LTE Handover (FAJ 121 0489)	Conflicting		All handovers disabled.
LTE Service-Triggered Outgoing NR IRAT Mobility (FAJ 121 5744)	Conflicting	Partially impairing	The LTE Service-Triggered Outgoing NR IRAT Mobility feature is disabled if the following conditions are fulfilled: – The Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE. – The value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to `QCI`. – The value of the `EUtranCellFDD.servOrPrioIFHoSetupBearer` or the `EUtranCellTDD.servOrPrioIFHoSetupBearer` attribute is set to `false`.
Outgoing NR IRAT Handover (FAJ 121 5377)	Conflicting	Partially impairing	If the Outgoing NR IRAT Handover feature is active and Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE, mobility to NR Standalone is blocked regardless of the UE reporting good or poor coverage.

Hardware

No specific hardware requirements.

Limitations

This feature can affect other features which configure measurements in the UE and trigger a handover to a reported cell. The following handovers are affected:

- Intra-frequency handovers
- Handovers to other LTE frequencies
- Handovers to other RATs

To avoid voice service delays caused by handovers, these handovers are disabled when all the following conditions are true:

- Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE
- `EUtranCellFDD.servOrPrioTriggeredIFHo` or `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to `QCI`
- `EUtranCellFDD.servOrPrioIFHoSetupBearer` or `EUtranCellFDD.servOrPrioIFHoSetupBearer` attribute is set to `false`

Network Requirements

No specific network requirements.

Related concepts
Engineering Guidelines for Service or Priority-Triggered Inter-Frequency Handover

3 Feature Operation of Service or Priority-Triggered Inter-Frequency Handover

> The feature can be configured by attributes to act on bearers with QCI=1 service level or bearers with a specific priority.

The effect of the Service or Priority-Triggered Inter-Frequency Handover feature is controlled by the `EUtranCellFDD.servOrPrioTriggeredIFHo`, and the `EUtranCellTDD.servOrPrioTriggeredIFHo` attributes.

The following values are possible:

- If the value is set to `NOT_ACTIVATED`, the feature is not activated in this cell.
- If the value is set to ARP, the behavior is always to set up bearers before handover.
- If the value is set to `QCI`, the feature can be configured to use one of two behaviors depending on the value of the `EUtranCellFDD.servOrPrioIFHoSetupBearer`, or `EUtranCellTDD.servOrPrioIFHoSetupBearer` attribute:
  - If the value is set to `true`, the behavior is to set up bearers before handover.
  - If the value is set to `false`, the behavior is to handover before setting up bearers.

Related Information
Service or Priority-Triggered Inter-Frequency Handover with Bearer Setup before Handover
Service or Priority-Triggered Inter-Frequency Handover with Handover before Bearer Setup

3.1 Service or Priority-Triggered Inter-Frequency Handover with Bearer Setup before Handover

> The feature is configurable to set up bearers before handover.

The feature can be triggered by the following procedures:

- E-RAB setup request from the MME
- Incoming handover
- RRC connection re-establishment
- Initial context setup

Operational Process

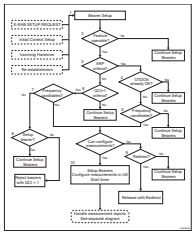

Figure 1 E-RAB Setup Request Handling, Bearer Setup before Handover

1. A set of bearers is set up for a UE. This occurs as part of the following procedures:

E-RAB SETUP REQUEST	A message from the MME which requests the eNodeB to set up a set of bearers for the UE. For more information, see *3GPP TS 36.413*, and Multiple Radio Bearers per User.
Incoming Handover	During incoming handover preparation, resources for a set of bearers are allocated in the eNodeB. During handover execution, the bearers are configured in the UE.
RRC Connection Re-establishment	When a UE initiates an RRC re-establishment, the bearers of the UE are set up in the UE again.
Initial Context Setup	At Initial Context Setup, a set of bearers is set up for the UE.

2. If this feature, or the Coverage-Triggered Inter-Frequency Handover feature is inoperable, the bearers are set up without being affected by this feature.
3. The bearer setup request is considered to fulfill the Allocation and Retention Priority (ARP) if the following conditions are met:
   - The `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` MO attribute has the value ARP `[1]`
   - The set of bearers contains at least one bearer with an ARP level that corresponds to a TRUE value in the `EUtranCellFDD.arpPriorityLevelForSPIFHo` or the `EUtranCellTDD.arpPriorityLevelForSPIFHo` MOM attribute.
4. The eNodeB can be configured so that the UE is only repositioned if there is an improvement of OTDOA. For more information, see OTDOA User Plane Location Support, and Inter-Frequency OTDOA. If the following criteria are fulfilled, the UE can stay in the current cell and the bearers can be set up without being affected by the Service or Priority-Triggered Inter-Frequency Handover feature:

   The `EUtranCellFDD.otdoaCheckEnabled` or `EUtranCellTDD.otdoaCheckEnabled` attribute is set to `true`, and at least one of the following criteria is true:
   - The UE does not support OTDOA. That is, the UE capability IE `otdoa-UE-Assisted-r10` is not present.
   - The UE supports OTDOA, and there is a PRS signal in the source cell. That is, the `otdoa-UE-Assisted-r10` UE capability IE is set to `supported`, and

the OTDOA User Plane Location Support feature is operable.

Also, the EUtranCellFDD.otdoaSuplActive or the EUtranCellTDD.otdoaSuplActive attribute is set to true for the cell.

– The UE and the eNodeB support Inter-Frequency OTDOA measurements. That is, the interFreqRSTD-Measurement-r10 UE capability IE is set to supported, and the Inter-Frequency OTDOA feature is operable.

5. The eNodeB determines the set of frequency candidates where the UE looks for possible handover target cells when the ARP of the bearer triggers this feature. This is done by including each EUtranFreqRelation that fulfills all the following criteria:
   – According to the capabilities of the UE, as reported to the eNodeB in the information element UE-EUTRA-Capability, the UE is able to operate on this frequency. When the Multiple Frequency Band Indicators feature is operable, more frequencies can be supported by the UE.
   – The EUtranFreqRelation.arpPrio attribute of the frequency states can be measured. A priority value of −1 means that the frequency is not measured.
   – The EUtranFreqRelation.mobilityAction attribute has the value HANDOVER.
   – If the frequency has a different mode (for example TDD for FDD), then the feature Intra-LTE Inter-Mode Handover must be operable.
   – If the Shared LTE RAN feature is operable, more selection criteria are used.
   – If the Subscriber Triggered Mobility feature is operable, more selection criteria are used.

   From this set of frequency candidates, the eNodeB selects the frequency relations with the highest value for arpPrio.

   Note: If several frequency relations share the highest value for this attribute, then they are all selected.

6. The bearer setup request is considered to fulfill the QCI criteria if:
   – The EUtranCellFDD.servOrPrioTriggeredIFHo or the EUtranCellTDD.servOrPrioTriggeredIFHo MO attribute has the value QCI [0].
   – The set of bearers contains at least one bearer with a QCI value equal to 1.

   If the request does not fulfill the QCI criteria, the bearers are set up without being affected by this feature.

7. The eNodeB determines the set of frequency candidates where the UE looks for possible handover target cells when the QCI of the bearer triggers this feature. This is done by including each EUtranFreqRelation that fulfills the following criteria:
   – According to the capabilities of the UE, as reported to the eNodeB in the information element UE-EUTRA-Capability, the UE is able to operate on this frequency. When the Multiple Frequency Band Indicators feature is operable, more frequencies can be supported by the UE.
   – The EUtranFreqRelation.voicePrio attribute of the frequency states can be measured. A priority value of −1 means that the frequency is not measured.
   – The EUtranFreqRelation.mobilityAction attribute is set to HANDOVER.
   – If the target frequency has a different mode (for example TDD for FDD), the feature Intra-LTE Inter-Mode Handover must be operable.
   – If the Shared LTE RAN feature is operable, more selection criteria are used.
   – If the Subscriber Triggered Mobility feature is operable, more selection criteria are used.
   – From this set of frequency candidates, the eNodeB selects the frequency relations with the highest value for EUtranFreqRelation.voicePrio. If several frequency relations share the highest value for this attribute, then they are all selected.

8. The feature is triggered by a bearer with QCI 1. Also, there are no frequency candidates where the UE can find a suitable target cell for handover. In this case, the behavior of the eNodeB differs depending on the value of the following MOM parameters:
   – EUtranCellFDD.servOrPrioTriggeredErabAction or EUtranCellTDD.servOrPrioTriggeredErabAction
     - If the value is SETUP all the bearers in the request are set up in the regular way.
     - If the value is REJECT the behavior depends on what triggered the bearer setup:

E-RAB SETUP REQUEST	All bearers with QCI=1 from the request are rejected in the eNodeB and included in the E-RAB Failed to Setup List with the cause Not allowed QCI value. Any other bearer in the request is set up and included in the E-RAB Setup List. The S1AP:E-RAB SETUP RESPONSE message is sent to the MME. **Note:** The E-RAB Failed to Setup List and the E-RAB Setup List IEs are included in the S1AP:E-RAB SETUP RESPONSE message.
Initial Context Setup	All bearers with QCI=1 from the request are rejected in the eNodeB and included in the E-RAB Failed to Setup List with the cause Not allowed QCI value. Any other bearer in the request is set up and included in the E-RAB Setup List. The S1AP:INITIAL CONTEXT SETUP RESPONSE message is sent to the MME. **Note:** The E-RAB Failed to Setup List and the E-RAB Setup List IEs are included in the S1AP:INITIAL CONTEXT SETUP RESPONSE message.
Incoming Handover Preparation	All bearers in the request are rejected by the feature. As a result, the X2:HANDOVER PREPARATION FAILURE or the S1AP:HANDOVER PREPARATION FAILURE response message is sent to the MME or Source eNodeB.
RRC Connection Re-establishment	All bearers with QCI=1 from the request are rejected. This causes the re-establishment to fail and a RRCConnectionReestablishmentReject message is sent to the UE.

9. There are frequency candidates which the UE can search for handover target cells, but it is not possible to configure any new measurements in the UE. In this case, the behavior depends on the value of the following MOM attributes:

- `EUtranCellFDD.servOrPrioRedirectEnabled`
- `EUtranCellTDD.servOrPrioRedirectEnabled`
-
  - If the attribute is set to FALSE, all the bearers in the request are set up in the regular way.
  - If the attribute is set to TRUE, the behavior depends on what triggered the bearer setup:
  -

E-RAB SETUP REQUEST	The `S1AP:E-RAB SETUP RESPONSE` response message is sent to the MME, with all bearers from the request included in the `E-RAB Failed to Setup List`. In `E-RAB Failed to Setup List` where the bearers with QCI=1 (see step 5) have the cause value `Not allowed QCI value`, and all other bearers in the request have an unspecified cause. This includes all bearers with a specific ARP level, see step 3. Then, the UE is released with a redirect to one of the frequencies selected in step 5 or step 7, using the release cause value `Release due to E-UTRAN Generated Reason`.
Initial Context Setup	The `S1AP:INITIAL CONTEXT SETUP FAILURE` response message is sent to the MME. The UE is released with a redirect to one of the frequencies selected in step 5 or step 7, using the release cause value `Release due to E-UTRAN Generated Reason`.
Incoming Handover Preparation	Setup of the bearers fails. This causes the handover preparation to fail.
RRC Connection Re-establishment	Setup of the bearers fails, and as a consequence, the eNodeB rejects the re-establishment.

10. The bearers included in the request are configured, both in the eNodeB and in the UE. Event A5 measurements are configured to find a suitable handover target cell.

    The behavior at Incoming Handover is slightly different. The `UeMeasControl.servOrPrioIFHoTimer` timer is started. Depending on what started the bearer setup procedure, the following happens:

E-RAB SETUP REQUEST	The `S1AP:E-RAB SETUP RESPONSE` response message is sent.
Initial Context Setup	No action is taken.
Incoming Handover Preparation	The bearers included in the request are prepared in the eNodeB. The UE is configured with the bearers when it connects to the eNodeB during handover execution. This also applies when the Event A5 measurements are configured in the UE.
RRC Connection Re-establishment	No action is taken.

Related Information
Feature Operation of Service or Priority-Triggered Inter-Frequency Handover
Event A5 Measurements

3.2 Service or Priority-Triggered Inter-Frequency Handover with Handover before Bearer Setup

> The feature is configurable for handover before bearer setup.

The feature can be triggered by the following procedures:

- E-RAB setup request from the MME
- Incoming handover
- RRC connection re-establishment
- Initial context setup

Operational Process

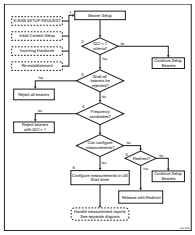

Figure 2 E-RAB Setup Request Handling, Handover before Bearer Setup

1. A set of bearers is set up for a UE. This occurs as part of the following procedures:

E-RAB SETUP REQUEST	A message from the MME which requests the eNodeB to set up a set of bearers for the UE, see *3GPP TS 36.413*, and Multiple Radio Bearers per User.
Incoming Handover	During incoming handover preparation, resources for a set of bearers are allocated in the eNodeB. During handover execution, the bearers are configured in the UE.
RRC Connection Re-establishment	When a UE initiates an RRC re-establishment, the bearers of the UE are set up in the UE again.
Initial Context Setup	At Initial Context Setup, a set of bearers is set up for the UE.

2. The bearer setup request is considered to fulfill the QCI criteria if the set of bearers contains at least one bearer with QCI=1.

   Note: If the request does not fulfill the QCI criteria, the bearers are set up without being affected by this feature.

3. All bearers in setup request can be rejected depending on what triggered the bearer setup:

E-RAB SETUP REQUEST	No bearers are rejected.
Incoming Handover Preparation	All bearers in the request are rejected by the feature. As a result, the `X2:HANDOVER PREPARATION FAILURE` or the `S1AP:HANDOVER PREPARATION FAILURE` response message is sent to the MME or the Source eNodeB.
RRC Connection Re-establishment	All bearers with QCI=1 from the request are rejected. As a result, the re-establishment fails and the `RRCConnectionReestablishmentReject` message is sent to the UE.
Initial Context Setup	No bearers are rejected.

4. The eNodeB determines the set of frequency candidates where the UE looks for possible handover target cells. This is done by including each

EUtranFreqRelation that fulfills the following criteria:

- According to the capabilities of the UE, as reported to the eNodeB in the information element UE-EUTRA-Capability, the UE is able to operate on this frequency. When the Multiple Frequency Band Indicators feature is operable, more frequencies can be supported by the UE.
- The EUtranFreqRelation.voicePrio attribute of the frequency states if it can be measured. A priority value of −1 means that the frequency is not measured.
- The EUtranFreqRelation.mobilityAction attribute is set to HANDOVER.
- If the target frequency has a different mode (TDD or FDD), the feature Intra-LTE Inter-Mode Handover must be operable.
- If the Shared LTE RAN feature is operable, more selection criteria are used.
- If the Subscriber Triggered Mobility feature is operable, more selection criteria are used.

From this set of frequency candidates, the eNodeB selects the frequency relations with the highest value for EUtranFreqRelation.voicePrio.

Note: If several frequency relations share the highest value for this attribute, they are all selected.

a. If there are no frequency candidates where the UE can find a suitable target cell for handover. In this case, the behavior depends on what triggered the bearer setup:

E-RAB SETUP REQUEST	All bearers with QCI=1 from the request are rejected in the eNodeB and included in the E-RAB Failed to Setup List with the cause Not allowed QCI value. Any other bearer in the request is set up and included in the E-RAB Setup List. The S1AP:E-RAB SETUP RESPONSE message is sent to the MME. Note: The E-RAB Failed to Setup List and the E-RAB Setup List IEs are included in the S1AP:E-RAB SETUP RESPONSE message.
Incoming Handover Preparation	Handover already rejected, see step 3.
RRC Connection Re-establishment	Re-establishment already rejected, see step 3.
Initial Context Setup	All bearers with QCI=1 from the request are rejected in the eNodeB and included in the E-RAB Failed to Setup List with the cause Not allowed QCI value. Any other bearer in the request is set up and included in the E-RAB Setup List. The S1AP:INITIAL CONTEXT SETUP RESPONSE message is sent to the MME. Note: The E-RAB Failed to Setup List and the E-RAB Setup List IEs are included in the S1AP:INITIAL CONTEXT SETUP RESPONSE message.

5. If there are frequency candidates which the UE can use to search for handover target cells, but it is not possible to configure any new measurements in the UE, the behavior depends on the value of the MOM attributes:

- EUtranCellFDD.servOrPrioRedirectEnabled
- EUtranCellTDD.servOrPrioRedirectEnabled

a. If EUtranCellFDD.servOrPrioRedirectEnabled or EUtranCellFDD.servOrPrioRedirectEnabled is false, the behavior depends on what triggered the bearer setup:

E-RAB SETUP REQUEST	All bearers with QCI=1 from the request are rejected in the eNodeB and included in the E-RAB Failed to Setup List with the cause Not allowed QCI value. Any other bearer in the request is set up and included in the E-RAB Setup List. The S1AP:E-RAB SETUP RESPONSE message is sent to the MME. Note: The E-RAB Failed to Setup List and the E-RAB Setup List IEs are included in the S1AP:E-RAB SETUP RESPONSE message.
Incoming Handover Preparation	Handover already rejected, see step 3.
RRC Connection Re-establishment	Re-establishment already rejected, see step 3.
Initial Context Setup	E-RAB Failed to Setup ListAll bearers with QCI=1 from the request are rejected in the eNodeB and included in the with the cause Not allowed QCI value. Any other bearer in the request is set up and included in the E-RAB Setup List. The S1AP:INITIAL CONTEXT SETUP RESPONSE message is sent to the MME. Note: The E-RAB Failed to Setup List and the E-RAB Setup List IEs are included in the S1AP:INITIAL CONTEXT SETUP RESPONSE message.

b. If EUtranCellFDD.servOrPrioRedirectEnabled or EUtranCellFDD.servOrPrioRedirectEnabled is true, the behavior depends on what triggered the bearer setup:

E-RAB SETUP REQUEST	In the eNodeB, all bearers from the request included in the E-RAB Failed to Setup List with QCI=1 (see step 2) have the cause value Not allowed QCI value, and all other bearers in the request have an unspecified cause. The S1AP:E-RAB SETUP RESPONSE response message is sent to the MME. The UE is released with a redirect to one of the frequencies selected in step 5 or step 4, using the release cause value Release due to E-UTRAN Generated Reason.
Incoming Handover Preparation	Handover already rejected, See step 3.

RRC Connection Re-establishment	Re-establishment already rejected, See step 3.
Initial Context Setup	The `S1AP:INITIAL CONTEXT SETUP FAILURE` response message is sent to the MME. The UE is released with a redirect to one of the frequencies selected in step 5 or step 4, using the release cause value `Release due to E-UTRAN Generated Reason`.

6. The bearers included in the request which have QCI=1 can be rejected depending on what started the bearer setup procedure:

E-RAB SETUP REQUEST	In the eNodeB, all bearers with QCI=1 from the request are rejected and included in the `E-RAB Failed to Setup List` with the cause `S1 intra system handover triggered`. Any other bearer in the request is set up. The `S1AP:E-RAB SETUP RESPONSE` response message is sent to the MME.
Incoming Handover Preparation	Handover is already rejected. See Step 3.
RRC Connection Re-establishment	Re-establishment already rejected. See Step 3.
Initial Context Setup	– If `EUtranCellTDD.spifhoSetupBearerAtInitialCtxtSetup` or `EUtranCellFDD.spifhoSetupBearerAtInitialCtxtSetup` is `true`, then all bearers in the request are set up. – If `EUtranCellTDD.spifhoSetupBearerAtInitialCtxtSetup` or `EUtranCellFDD.spifhoSetupBearerAtInitialCtxtSetup` is `false`, then all bearers with QCI=1 from the request are rejected in the eNodeB and included in the `E-RAB Failed to Setup List` with the cause `Not allowed QCI value`. All other bearers in the request are set up. The `S1AP:E-RAB SETUP RESPONSE` response message is sent to the MME.

The remaining bearers included in the request are configured, both in the eNodeB and in the UE. Also, Event A5 measurements are configured to find a suitable handover target cell, see Event A5 Measurements. Also, the timer `UeMeasControl.servOrPrioIFHoTimer` is started.

Related Information
Feature Operation of Service or Priority-Triggered Inter-Frequency Handover
Event A5 Measurements

3.3 Event A5 Measurements

> To find a target cell for handover, the eNodeB sets up Event A5 measurements in the UE.

The configuration of the Event A5 measurements is based on the attributes in the `ReportConfigA5` or the `ReportConfigA5Spifho`MO. The`ReportConfigA5` MO is also used when the eNodeB configures the Event A5 measurements for the Coverage-Triggered Inter-Frequency Session Continuity and the Mobility Control at Poor Coverage features.

The `EUtranCellFDD.spifhoDedicatedReportConfig` or the `EUtranCellFDD.spifhoDedicatedReportConfig` MO can be used to enable Event A5 measurement configuration for the Service or Priority-Triggered Inter-Frequency Handover feature.

If the value of the `EUtranCellFDD.spifhoDedicatedReportConfig` or the `EUtranCellFDD.spifhoDedicatedReportConfig` attribute is set to `false`, the following MO attributes are used for this feature:

- `ReportConfigA5.a5Threshold2Rsrp`
- `ReportConfigA5.hysteresisA5`
- `ReportConfigA5.timeToTriggerA5`

In this case, the trigger quantity is always the RSRP. `Threshold1` is hard-coded to -44 dBm to ensure that *Entering condition 1* is always fulfilled for Event A5. For more information, see *3GPP TS 36.331*.

If the Service Triggered Mobility feature is operable, the value of the `a5Threshold2RsrpOffset` member of the `QciProfileOperatorDefined.measReportConfigParams` and the `QciProfilePredefined.measReportConfigParams` attributes can be used. If the Multi-Layer Service-Triggered Mobility feature is operable, the value of the `EUtranFreqRelation.a5Thr2RsrpFreqOffset` attribute can be used.

If the value of the `EUtranCellFDD.spifhoDedicatedReportConfig` or the `EUtranCellFDD.spifhoDedicatedReportConfig` attribute is set to `true`, the feature-specific A5 Event measurements can be configured with the `ReportConfigA5Spifho` MO. In this case, the following additional options are available:

- RSRQ can be set as a trigger quantity.
- `Threshold1` is configurable with the `ReportConfigA5Spifho.a5Threshold1Rsrp` and the `ReportConfigA5Spifho.a5Threshold1Rsrq` attributes.

Both RSRP and RSRQ can be checked for the Service or Priority-Triggered Inter-Frequency Handover feature-specific event A5 measurements regardless of the value of the `EUtranCellFDD.spifhoDedicatedReportConfig` or the `EUtranCellFDD.spifhoDedicatedReportConfig` attribute. If the `UeMeasControl.bothA5RsrpRsrqCheckSpifho` attribute is set to true, handover occurs only if conditions are met both for the RSRP_target and the RSRQ_target.

Related Information
Service or Priority-Triggered Inter-Frequency Handover with Bearer Setup before Handover

Service or Priority-Triggered Inter-Frequency Handover with Handover before Bearer Setup

3.3.1 Handling of Event A5 Measurement Reports

> Event A5 Measurement reports are used to determine if handover is possible for the UE.

The handling of Event A5 measurement reports is illustrated in Figure 3.

Operational Process

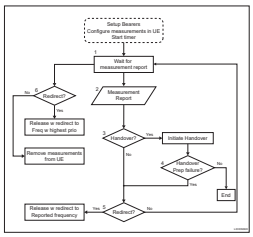

Figure 3 Operational Sequence Diagram

An E-RAB has been set up in the UE, which requires the UE to be handed over to another cell. Event A5 measurements have been configured in the UE to search for such a cell. The timer, `UeMeasControl.servOrPrioIFHoTimer`, has been started to handle the case when a cell is reported by the UE.

1. The eNodeB waits for a measurement report from the Event A5 measurements configured in 6.
2. An Event A5 measurement report arrives from the UE.
3. If the measurement report includes a valid handover target cell, a handover is initiated. A cell is valid for handover when all the following criteria are fulfilled:
   - The reported cell is known in the source cell. That is, a `EUtranCellRelation` exists to the reported cell from the cell where the UE is connected.
   - Handover is allowed to the cell according to the MOM attribute `EUtranCellRelation.isHoAllowed`.
   - The UE FGI bit 13, Inter-Frequency Handover within FDD or TDD, is set to 1.
   - If the frequency band of the target cell has a different mode (for example TDD for FDD), the UE FGI bit 30, Handover between FDD and TDD, is set to 1.
4. When the eNodeB determines that no handover can be initiated based on the reported cells, the behavior depends on the value of the MOM attribute `EUtranCellFDD.servOrPrioRedirectEnabled` or `EUtranCellTDD.servOrPrioRedirectEnabled`:

   If the value is `true`, the eNodeB releases the UE with a redirection to the frequency of the reported cell using the cause value `User Inactivity`.

   If the value is `false`, the eNodeB does nothing and waits for another measurement report.
5. The timer `UeMeasControl.servOrPrioIFHoTimer` expires. The behavior depends on the value of the MOM attribute `EUtranCellFDD.servOrPrioRedirectEnabled` or `EUtranCellTDD.servOrPrioRedirectEnabled`.

   If the value is `true`, the eNodeB releases the UE with a redirection to the frequency with the highest priority. The release cause value `Release due to E_UTRAN Generated Reason` is used, based on the same priority attribute used when selecting the measured frequencies. If the value is `false`, the eNodeB removes the configured Event A5 measurements from the UE and keeps it in the current cell.

   Note: If `UeMeasControl.servOrPrioIFHoTimer` has the value 0, the timer never expires.

4 Network Impact for Service or Priority-Triggered Inter-Frequency Handover

> The feature has no impact on the network.

Capacity and Performance

No impact.

Interfaces

No impact.

Other Network Elements

No impact.

5 Parameters for Service or Priority-Triggered Inter-Frequency Handover

The feature introduces MOM attributes for configuration, and affects already existing attributes related to Event A5.

Table 2 Parameters

Parameter	Type	Description
EUtranCellFDD.servOrPrioTriggeredIFHo	Introduced	See MOM description.
EUtranCellTDD.servOrPrioTriggeredIFHo	Introduced	See MOM description.
EUtranCellFDD.otdoaCheckEnabled	Introduced	See MOM description.
EUtranCellTDD.otdoaCheckEnabled	Introduced	See MOM description.
EUtranCellFDD.servOrPrioIFHoSetupBearer	Introduced	Controls the behavior of the eNodeB when the Service or Priority-Triggered Inter-Frequency Handover feature is triggered by a bearer with QCI=1. It gives the possibility to configure for handover to target cell before allowing QCI=1 bearer setup.
EUtranCellTDD.servOrPrioIFHoSetupBearer	Introduced	Controls the behavior of the eNodeB when the Service or Priority-Triggered Inter-Frequency Handover feature is triggered by a bearer with QCI=1. It gives the possibility to configure for handover to target cell before allowing QCI=1 bearer setup.
EUtranCellFDD.spifhoSetupBearerAtInitialCtxtSetup	Introduced	Controls if a bearer with QCI=1, handled by the Service or Priority-Triggered Inter-Frequency Handover feature, will be set up at Initial Context Setup even when the attribute `servOrPrioIFHoSetupBearer` is false.
EUtranCellTDD.spifhoSetupBearerAtInitialCtxtSetup	Introduced	Controls if a bearer with QCI=1, handled by the Service or Priority-Triggered Inter-Frequency Handover feature, will be set up at Initial Context Setup even when the attribute `servOrPrioIFHoSetupBearer` is false.
EUtranCellFDD.servOrPrioRedirectEnabled	Introduced	See MOM description.
EUtranCellTDD.servOrPrioRedirectEnabled	Introduced	See MOM description.
EUtranFreqRelation.arpPrio	Introduced	See MOM description.
UeMeasControl.servOrPrioIFHoTimer	Introduced	See MOM description.
EUtranCellFDD.servOrPrioTriggeredErabAction	Introduced	Controls the behavior of the eNodeB when the Service or Priority-Triggered Inter-Frequency Handover feature is triggered by a bearer with QCI=1, and there are no frequency candidates where the UE can find a suitable target cell for handover. If the value is SETUP, all bearers in the request are set up in the regular way. If the value is REJECT, the bearer or bearers with QCI=1 are rejected with the cause value `Not allowed QCI value`.
EUtranCellTDD.servOrPrioTriggeredErabAction	Introduced	Controls the behavior of the eNodeB when the Service or Priority-Triggered Inter-Frequency Handover feature is triggered by a bearer with QCI=1, and there are no frequency candidates where the UE can find a suitable target cell for handover. If the value is SETUP, all bearers in the request are set up in the regular way. If the value is REJECT, the bearer or bearers with QCI=1 are rejected with the cause value `Not allowed QCI value`.
ReportConfigA5Spifho.a5Threshold1Rsrp	Introduced	See MOM description.
ReportConfigA5Spifho.a5Threshold1Rsrq	Introduced	See MOM description.
ReportConfigA5Spifho.a5Threshold2Rsrp	Introduced	See MOM description.
ReportConfigA5Spifho.a5Threshold2Rsrq	Introduced	See MOM description.
ReportConfigA5Spifho.hysteresisA5	Introduced	See MOM description.
ReportConfigA5Spifho.reportConfigA5SpifhoId	Introduced	See MOM description.
ReportConfigA5Spifho.timeToTriggerA5	Introduced	See MOM description.
ReportConfigA5Spifho.triggerQuantityA5	Introduced	See MOM description.
EUtranCellFDD.spifhoDedicatedReportConfig	Introduced	See MOM description.
EUtranCellTDD.spifhoDedicatedReportConfig	Introduced	See MOM description.
UeMeasControl.bothA5RsrpRsrqCheckSpifho	Introduced	See MOM description.

Parameter	Type	Description
EUtranFreqRelation.voicePrio	Affecting	See MOM description.
ReportConfigA5.a5Threshold2Rsrp	Affecting	See MOM description.
ReportConfigA5.hysteresisA5	Affecting	See MOM description.
ReportConfigA5.timeToTriggerA5	Affecting	See MOM description.

Related Information

Service or Priority-Triggered Inter-Frequency Handover with Bearer Setup before Handover
Service or Priority-Triggered Inter-Frequency Handover with Handover before Bearer Setup

6 Performance of Service or Priority-Triggered Inter-Frequency Handover

> The feature is associated with counters for measuring attempted and successful handovers.

KPIs

This feature has no associated KPIs.

Counters

- EUtranCellFDD.pmErabEstabAttAddedQci
- EUtranCellFDD.pmErabEstabSuccAddedQci
- EUtranCellFDD.pmErabEstabAttInitQci
- EUtranCellFDD.pmErabEstabSuccInitQci
- EUtranCellTDD.pmErabEstabAttAddedQci
- EUtranCellTDD.pmErabEstabSuccAddedQci
- EUtranCellTDD.pmErabEstabAttInitQci
- EUtranCellTDD.pmErabEstabSuccInitQci
- EUtranFreqRelation.pmSpifhoMeasuredUe
- EUtranCellRelation.pmHoPrepAttLteSpifho
- EUtranCellRelation.pmHoPrepSuccLteSpifho
- EUtranCellRelation.pmHoExeAttLteSpifho
- EUtranCellRelation.pmHoExeSuccLteSpifho
- EUtranCellRelation.pmHoPrepAttNonMob
- EUtranCellRelation.pmHoPrepSuccNonMob
- EUtranCellRelation.pmHoExeAttNonMob
- EUtranCellRelation.pmHoExeSuccNonMob
- EUtranCellRelation.pmHoPrepAttLteInterF
- EUtranCellRelation.pmHoExeAttLteInterF
- EUtranCellRelation.pmHoPrepSuccLteInterF
- EUtranCellRelation.pmHoExeSuccLteInterF
- EUtranCellRelation.pmHoPrepSuccLteInterF

Events

Table 3 Events

Event	Event Parameter	Description
INTERNAL_PROC_HO_PREP_S1_OUT	EVENT_PARAM_PROC_HO_PREP_OUT_ATTEMPT_CAUSE	The cause of the procedure for Handover Preparation. Circuit Switched, Carrier Aggregation, Service or Priority-triggered Inter-frequency Handover. When the handover is triggered by the feature Service or Priority-triggered Inter-frequency Handover, the value is EVENT_VALUE_SPIFHO.
INTERNAL_PROC_HO_PREP_X2_OUT	EVENT_PARAM_PROC_HO_PREP_OUT_ATTEMPT_CAUSE	The cause of the procedure for HO Preparation. Circuit Switched, Carrier Aggregation, Service or Priority-triggered Inter-frequency Handover. When the handover is triggered by the feature Service or Priority-triggered Inter-frequency Handover, the value is EVENT_VALUE_SPIFHO.
INTERNAL_PROC_HO_EXEC_S1_OUT	EVENT_PARAM_PROC_HO_EXEC_OUT_ATTEMPT_CAUSE	The cause of the procedure for HO Preparation. Circuit Switched, Carrier Aggregation, Service or Priority-triggered Inter-frequency Handover. When the handover is triggered by the feature Service or Priority-triggered Inter-frequency Handover, the value is EVENT_VALUE_SPIFHO.
INTERNAL_PROC_HO_EXEC_X2_OUT	EVENT_PARAM_PROC_HO_EXEC_OUT_ATTEMPT_CAUSE	The cause of the procedure for HO Preparation. Circuit Switched, Carrier Aggregation, Service or Priority-triggered Inter-frequency Handover. When the handover is triggered by the feature Service or Priority-triggered Inter-frequency Handover, the value is EVENT_VALUE_SPIFHO.

Event	Event Parameter	Description
INTERNAL_EVENT_UE_MOBILITY_EVAL	EVENT_PARAM_MOBILITY_TRIGGER	Indicates the trigger for performing a mobility evaluation. EVENT_VALUE_MEAS_REPORT_A5_SPIFHO, EVENT_VALUE_SPIFHO
INTERNAL_EVENT_UE_MOBILITY_EVAL	EVENT_PARAM_MOBILITY_DECISION	Indicates the decision the mobility evaluation has taken. Service or Priority-triggered Inter-frequency Handover. EVENT_VALUE_RECONFIGURE_MEASUREMENTS_SPIFHO

7 Activate Service or Priority-Triggered Inter-Frequency Handover

> To use the feature, it must be activated following the usual feature activation procedure.

Prerequisites

- The license key is installed in the node.
- CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `FeatureState.featureState` attribute to `ACTIVATED` in the `FeatureState=CXC4011736` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

8 Deactivate Service or Priority-Triggered Inter-Frequency Handover

> If the feature is no longer needed, it can be deactivated following the usual feature deactivation procedure. It must also be deactivated before the activation of any conflicting feature.

Prerequisites

CCTR is active for at least one week before this procedure to collect enough troubleshooting data.

Steps

1. Set the `FeatureState.featureState` attribute to `DEACTIVATED` in the `FeatureState=CXC4011736` MO instance.

After This Task

Keep CCTR active for at least one week after this procedure for continued collection of troubleshooting data.

9 Engineering Guidelines for Service or Priority-Triggered Inter-Frequency Handover

> The Service or Priority-Triggered Inter-Frequency Handover feature can cause delays in the setup of voice service during UE handovers.

The following procedures can cause delays in the voice service:

- Intra-frequency handover
- Handover to other LTE frequencies
- Handover to other RATs

To avoid service delays, the handovers during these cases are disabled when the all the following conditions are met:

- Service or Priority-Triggered Inter-Frequency Handover measurements are configured in the UE.
- The value of the `EUtranCellFDD.servOrPrioTriggeredIFHo` or the `EUtranCellTDD.servOrPrioTriggeredIFHo` attribute is set to `QCI`
- The value of the `EUtranCellFDD.servOrPrioIFHoSetupBearer` or the `EUtranCellFDD.servOrPrioIFHoSetupBearer` attribute is set to `false`

Feature-Specific Event A5 Measurement Configuration

An event A5 measurement configuration that is specific to the Service or Priority-Triggered Inter-Frequency Handover feature can be activated with the following attributes:

- `EUtranCellFDD.spifhoDedicatedReportConfig`
- `EUtranCellTDD.spifhoDedicatedReportConfig`

When a handover action that is triggered by the Service or Priority-Triggered Inter-Frequency Handover feature is evaluated, an A5 measurement report is sent. The value of the `UeMeasControl.bothA5RsrpRsrqCheckSpifho` attribute controls whether both RSRP and RSRQ are checked during handover evaluation.

Related reference
Dependencies of Service or Priority-Triggered Inter-Frequency Handover

10 Feature Change History

This section lists changes that affected this feature and the impact it had on the network.

10.1 21.Q3: Differentiated VoLTE Mobility

This feature enhancement introduces specific event A5 configurations for the Service or Priority-Triggered Inter-Frequency Handover feature.

Feature Name:	Service or Priority-Triggered Inter-Frequency Handover
Feature Identity:	FAJ 121 3087
Value Package Name:	Service-Based Mobility
Value Package Identity:	FAJ 801 0433
Node Type:	Baseband Radio Node
Access Type:	LTE

With the feature enhancement, an event A5 measurement configuration that is specific to the Service or Priority-Triggered Inter-Frequency Handover feature can be activated. To activate this configuration, the value of the `EUtranCellFDD.spifhoDedicatedReportConfig` or the `EUtranCellTDD.spifhoDedicatedReportConfig` attribute must be set to `true`.

The feature enhancement introduces a configurable solution for checking both RSRP and RSRQ in the event A5 measurement report for the Service or Priority-Triggered Inter-Frequency Handover feature. If the `UeMeasControl.bothA5RsrpRsrqCheckSpifho` attribute is set to `true`, handover occurs if the target conditions are met both for RSRP and RSRQ.

Benefits

With the introduction of dedicated attributes, the Service or Priority-Triggered Inter-Frequency Handover feature can be configured independently from other mobility features that use event A5 measurements.
The dedicated event A5 configuration brings the following benefits:

- The feature can be configured independently from other mobility features that use event A5 measurements.
- VoLTE mobility is enabled when the feature is configured to handover before bearer setup when no voice bearer is set up.
  Note: The feature is configured to handover before bearer setup when the `EUtranCellFDD.servOrPrioIFHoSetupBearer` or the `EUtranCellTDD.servOrPrioIFHoSetupBearer` attribute is set to `false`.
- The `timetotrigger` attribute can be configured independently. When the attribute is set to `0`, a VoLTE attempt can be redirected to FDD immediately. This can improve VoLTE setup time.
- `Threshold1` and the triggering quantity are configurable.

Capacity and Performance

With the feature-specific event A5 measurement configuration, the frequency with which the UE performs the mobility actions that are triggered by the Service or Priority-Triggered Inter-Frequency Handover feature can increase or decrease.
This feature enhancement can affect the following KPIs:

- Mobility KPIs
- Accessibility KPIs
- Retainability KPIs

Operation

The feature enhancement introduces the `ReportConfigA5Spifho` MO class.
The following attributes are introduced:

- `ReportConfigA5Spifho.a5Threshold1Rsrp`
- `ReportConfigA5Spifho.a5Threshold1Rsrq`
- `ReportConfigA5Spifho.a5Threshold2Rsrp`
- `ReportConfigA5Spifho.a5Threshold2Rsrq`
- `ReportConfigA5Spifho.hysteresisA5`
- `ReportConfigA5Spifho.reportConfigA5SpifhoId`
- `ReportConfigA5Spifho.timeToTriggerA5`
- `ReportConfigA5Spifho.triggerQuantityA5`
- `EUtranCellFDD.spifhoDedicatedReportConfig`
- `EUtranCellTDD.spifhoDedicatedReportConfig`
- `UeMeasControl.bothA5RsrpRsrqCheckSpifho`

The QCI offsets do not apply for the dedicated configuration.

Interfaces

No impact.

Hardware

No special hardware requirements.

Other Network Elements

ENM version 21.3 or later is required for this feature enhancement.

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